

Identification and Characterization of Pure Sleep Epilepsy in a Cohort of Patients With a First Seizure

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Abstract

Background and Objectives

Our aim was to study the development of pure sleep epilepsy after a first-ever seizure from sleep in adults.

Methods

This was a prospective observational study of patients seen at a tertiary hospital-based first seizure clinic between 2000 and 2011. Adults with a first-ever unprovoked seizure from sleep were consecutively recruited. All patients were followed up at least once after the initial seizure, and those not requiring regular clinical review were contacted every 1 to 2 years. The timing and pattern of subsequent seizures and potential predictors of future awake seizures were analyzed.

Results

Two hundred thirty-nine adults with a first-ever unprovoked seizure from sleep were identified. Sixty-one percent were male; mean age was 43 years (range 14–88 years); and median follow-up of 8.8 years (range 2 months–18 years). Of the 174 patients who had recurrent seizures, 130 patients (75%) had their second seizure from sleep, and of these, 76 of 94 (81%) also had their third seizure from sleep. Eighty-nine patients (37%) developed awake seizures during follow-up. In half of these patients, the awake seizure occurred within 2 years of the initial seizure. The risk of an awake seizure within 1 year of a first-ever seizure from sleep was 13.9% (95% CI 9.4%–18.3%), falling to 2.0% to 5.3% per year after 3 years. The risks of an awake seizure within 1 year of a second or third consecutive sleep seizure were 9.9% (95% CI 4.6%–15.3%) and 8.7% (95% CI 2.0%–15.4%), respectively, and similarly decreased with time.

Discussion

Most initial seizure recurrences after a first-ever sleep seizure occur during sleep. While more than one-third eventually had awake seizures, the annual risk of an awake seizure was $\leq 14\%$ and decreased with time, albeit with a small ongoing risk of between 2% and 5% per year. These findings may be used in counseling patients with seizures from sleep and to inform driving recommendations.

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Glossary

ASM = antiseizure medication; HR = hazard ratio; PSE = pure sleep epilepsy.

Pure sleep epilepsy (PSE) is defined as an “established pattern of seizures purely from sleep.”¹⁻³ However, the duration of epilepsy and number of sleep-only seizures required to make this diagnosis are unclear.¹⁻⁴ This has important management implications, including for driving.

Few studies have assessed the risk of future awake seizures in patients with sleep seizures.^{1,3-7} These studies have retrospectively diagnosed PSE, with variable inclusion criteria and definitions.^{1,3-7} The risk of awake recurrences ranges between 11% and 30%, with variable follow-up periods and relatively small sample sizes. Most awake seizures occur during the initial years after PSE diagnosis, but some occur decades later.¹⁻⁵ One study prospectively assessing the risk of awake seizures in established PSE found that only 11% of patients developed awake seizures during a variable follow-up of up to 6 years.⁴ This selected cohort may have a more benign course because the mean duration of epilepsy at enrollment was 54.6 months and most patients (78%) remained seizure-free for the first 2 years after enrollment.⁴ More frequent seizures, combination antiseizure medications (ASM), and discontinuation of ASMs are risk factors for awake seizures in PSE.^{1,3,4,6} However, these risk factors are not relevant to patients presenting with a first-ever sleep seizure. One study noted that children presenting with a first-ever seizure from sleep had a higher 2-year recurrence rate (53%) than those with their first seizure during wakefulness (30%), with most recurrences occurring during sleep.⁷ Another study confirmed that adults with a first-ever seizure from sleep were more likely to have a second seizure but did not state whether the recurrence was from sleep or wakefulness.⁸

In this study, by prospectively assessing the risk of developing awake seizures after a first seizure from sleep, we aimed to define the number of consecutive sleep seizures and the duration of recurrent sleep seizures required to more confidently diagnose PSE. We evaluated potential risk factors for subsequent awake seizures and information to counsel patients with seizures from sleep, including driving advice.⁶

Methods

The Western Australian first seizure database was established in 2000 and prospectively studies outcomes in adults with first seizures. The methodology has been described previously.⁹ All consecutive patients with an unprovoked first-ever seizure from sleep seen between 2000 and 2011 were included. Definition of a seizure from sleep was based on witness account when present and on patient recollection. Patients with a seizure on awakening, if they could recall waking first, even if still in bed, were regarded as having an awake seizure and

therefore excluded. Most (70%) patients were referred from emergency departments, with the remainder from general practitioners or other subspecialties. Patients who had a second seizure after being referred but before being seen at the clinic were included, with the initial seizure being analyzed as the first-ever seizure and the second defined as a recurrence. Clinical data were obtained at enrollment. Neuroimaging (CT or MRI) was obtained in all but 1 patient (CT and MRI in 167, CT only in 57, MRI only in 14). MRI scans were performed on a 1.0T Picker (Marconi Medical, Blackwood, NJ), 1.5T Siemens Sonata, or 1.5T Siemens Vision system (Siemens Healthineers, Erlangen, Germany) using multiplanar T1, T2, and fluid-attenuated inversion recovery whole-brain and dedicated temporal lobe imaging, with contrast administered as clinically indicated. Neuroimaging abnormalities were defined as epileptogenic if a lesion was identified relevant to the cause of seizure.¹⁰ A 21-channel EEG of 20 to 30 minutes' duration, usually with concurrent video, was performed in 97% of patients. The first EEG was analyzed and classified as being normal or abnormal, with abnormalities categorized as epileptiform or nonspecific.

Unprovoked seizures were defined as remote symptomatic with a history or neuroimaging evidence of a prior brain insult (including brain tumors) and idiopathic if there was no obvious cause (encompassing genetic epilepsy syndromes and cryptogenic seizures). Seizure type was determined by clinical assessment and according to published guidelines.^{11,12} Epilepsy type (electroclinical syndrome) was classified as focal, generalized, or unclassified on the basis of seizure semiology, EEG and neuroimaging findings, and other relevant medical history.¹² A positive family history was defined as a first-degree family member with a history of seizures. Decisions about ASM initiation after the first-ever seizure or recurrence were based on the clinician's opinion and patient factors. Patients treated after the first seizure were considered untreated if ASM was discontinued within 4 weeks. PSE was defined as ≥ 2 unprovoked seizures from sleep and without awake seizures during follow-up.

Follow-up

Patients were reviewed at clinic 3 to 9 months after the index seizure. If there were ongoing seizures, patients were followed up in the epilepsy clinic. If there were no further seizures, patients were discharged and contacted by phone every 1 to 2 years until seizure recurrence or death. For those not contactable by phone, the integrated state-based hospital computer system was checked for attendance at West Australian hospitals, and medical records were obtained to assess whether they had further seizures or died. Patients who developed a new CNS process after the first-ever seizure that

may have contributed to further seizures (e.g., stroke, significant head injury) were followed up until the time of CNS insult and then censored. All follow-up for this study ceased in April 2020.

Outcomes and Statistical Analysis

The primary outcome assessed was occurrence of an awake seizure, thereby defining the patient as not having PSE. Comparisons between groups were conducted with *t* tests for normally distributed data and Mann-Whitney or Kruskal-Wallis test for nonnormal data. The χ^2 and Fisher exact tests were used for categorical data. The Holm method was used to correct for multiple comparisons. Time to first awake seizure was analyzed with Kaplan-Meier survival curves and log-rank statistics, with point censoring of patients if an awake seizure had not occurred at the time of last follow-up or death. The cumulative and conditional (given survival up to a specific time) likelihoods of awake seizures and their CIs were calculated.

Potential risk factors for an awake seizure were examined with Cox proportional hazards models and included age, family history of epilepsy, seizure type, electroclinical syndrome, seizure etiology, time to recurrence, epileptogenic lesion on neuroimaging, epileptiform EEG abnormalities, and ASM treatment. Variables significant at the 10% level in preliminary univariate analyses were included in the multivariate models. Hazard ratios (HRs) and 95% CIs for multivariate analyses were calculated from the final models. Results were considered statistically significant at the 5% level.

Standard Protocol Approvals, Registrations, and Patient Consents

This study was approved by the Royal Perth Hospital Human Research Ethics Committee. Verbal or written consent was obtained from all patients on initial review.

Data Availability

Anonymized data not published within this article are available by request.

Results

During the 11 years of prospective recruitment, 239 patients with a first-ever unprovoked seizure from sleep were identified (61% male, mean age 43 years, range 14–88 years). Most presented with a tonic-clonic seizure (98%). Two hundred twenty patients (92%) had a witness to the first seizure, and 152 patients (63.6%) had orolingual trauma (112), incontinence (53), Todd paresis (15), and/or a seizure-related injury (21) alerting them to the seizure. The median time from first seizure to initial clinic review was 1 month (range 0–12 months). The median time to EEG was 15 days (90% within 4 months, range 0–14 months). The median duration of follow-up was 8.8 years (range 2 months–18 years; interquartile range 6.3 years). Seven patients (3%) had <1 year of follow-up (3 with 1 sleep seizure, 4 with recurrent sleep seizures).

Seizure Recurrence

Of 239 patients, 174 (72.8%) had a seizure recurrence, with 130 of the 174 (74.7%) having their second seizure from sleep (Figure 1). Most recurrences were tonic-clonic seizures (85.3%), accounting for 94% of second seizures from sleep and 61% of second seizures from wakefulness. After 2 consecutive sleep seizures, 76 of 94 (81%) with a third seizure also had their third seizure from sleep. Of the 239 patients, 89 (37%) with an initial seizure from sleep developed awake seizures. Their first awake seizure occurred as their second-ever seizure in 44 (49%) and as their third-ever seizure in 18 (20%) (Figure 1).

The cumulative risk of any seizure recurrence (sleep or awake) after a first-ever unprovoked seizure from sleep at 1 year was 54.8% (95% CI 48.5%–61.2%), at 2 years was 60.4% (95% CI 54.2%–66.7%), and at 10 years was 76.8% (95% CI 70.8%–82.9%) (Figure 2). The cumulative risk of an awake seizure at 1 year was 13.9% (95% CI 9.4%–18.3%), at 2 years was 19.3% (95% CI 14.2%–24.4%), and at 10 years was 42.1% (95% CI 34.8%–49.4%), with an ongoing annual conditional risk falling to 2.0% to 5.3% per year after 3 years (Figures 2 and 3). The risks of an awake seizure within 1 year of a second or third consecutive sleep seizure were lower at 9.9% (95% CI 4.6%–15.3%) and 8.7% (95% CI 2.0%–15.4%), respectively, with ongoing annual risks of awake seizures also between 2% and 5% (Figure 4). The median time until the first awake seizure was 2 years, with a wide range of 1 month to 18 years.

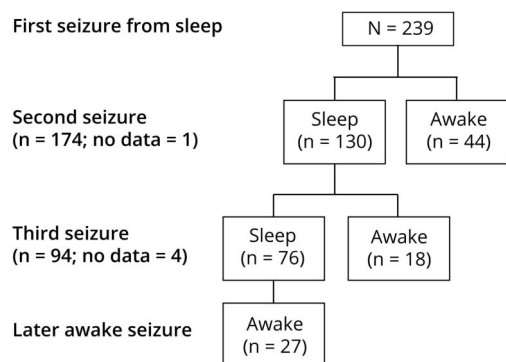
Exclusion of patients who had a seizure recurrence after being referred but before being seen in the clinic showed similar results, with the cumulative risk of an awake seizure at 1 year of 9.0% (95% CI 4.7%–13.4%) and similarly the ongoing annual conditional risk falling to 2.0% to 5.2% but after 2 rather than 3 years.

Clinical and Investigation Findings

The findings of patients with PSE (85 patients) and those with subsequent awake seizures are summarized in Table 1. Clinical features of both groups were similar. Follow-up duration was >2 years after the second seizure in 75 of 85 (88%) with PSE and 82 of 89 (92%) with subsequent awake seizures. Patients with PSE were less likely to have been treated after their first-ever seizure, and generalized epileptiform abnormalities were not seen in patients with PSE, but these differences were not significant after correction for multiple comparisons. There was no difference in mortality, including sudden unexpected death in epilepsy.

Univariate HRs for awake seizure recurrence are shown in Table 2. The only independent multivariate predictors of awake seizure occurrence were family history of epilepsy (HR 2.12, 95% CI 1.23–3.67; *p* = 0.007) and epileptiform EEG abnormalities (HR 1.89, 95% CI 1.14–3.13; *p* = 0.013). Only 10.8% of patients with PSE had epileptiform abnormalities on their first EEG, and all were focal epileptiform abnormalities. On multivariate analysis, age, seizure type, electroclinical

Figure 1 Flow Diagram of First Awake Seizure as Second, Third, or Subsequent Seizure After Initial Sleep Seizure (Patient Numbers)



syndrome, seizure clusters, time to recurrence, lesion on imaging, seizure etiology, and treatment after first seizure did not predict awake seizure occurrence. Treatment after first seizure also did not alter the likelihood of recurrent sleep seizures (HR 0.89, 95% CI 0.56–1.41; $p = 0.62$). The proportion treated at any stage during follow-up was 93% for those with PSE and 94% for those who developed awake seizures.

Discussion

Most seizure recurrences after a first-ever unprovoked seizure from sleep occurred during sleep. However, more than a third of patients eventually developed awake seizures, mostly within the first 3 years, with half of the awake recurrences being the second-ever seizure. The risk of awake seizures was

highest within the first year, with the annual conditional risk decreasing both with time and with recurrent sleep seizures. A 2% to 5% ongoing annual risk of awake seizures continued long term.

Our findings are similar to previous studies.^{1,3-5} Most report that 10% to 38% of patients with PSE develop awake seizures, mainly within the first 2 years^{4,5} and with a higher rate of focal seizures.^{1,3,4,7}

According to our survival analysis, of 100 patients presenting with a first-ever unprovoked sleep seizure, over the following 10 years, 23 patients will have no recurrence, 35 patients will develop PSE, and 42 patients will have an awake seizure. While the risk of awake seizures persists even after many years of multiple sleep-only seizures, this risk is low compared to risk in other patients with epilepsy.

The only predictors for awake seizures after a first-ever sleep seizure on multivariate analysis were a positive family history of epilepsy and EEG epileptiform abnormalities, especially generalized epileptiform abnormalities. However, even when combined, these predictors are insufficiently discriminative. Few patients had epileptiform abnormalities on their initial EEG, with most being focal, a finding similar to other first-seizure adult cohorts.^{1,3,4} All patients with recurrent seizures and generalized epileptiform abnormalities developed awake seizures. Neuroimaging findings did not predict awake seizures, consistent with previous studies,^{1,4} and treatment after first seizure did not alter the rate of seizure recurrence or of subsequent awake seizures.

Our data allow a more quantitative and prospective definition of PSE. We propose that the “established pattern of seizures purely from sleep” describing PSE can be defined as 3 consecutive seizures from sleep with no previous awake seizures or 3 consecutive years of sleep-only seizures when the annual conditional risk of an awake seizure has fallen to $\leq 5\%$. This is important for determining the risk of awake seizures during activities such as working with heavy machinery, participating in some sports and, most important, driving a motor vehicle.

Australian driving guidelines require 12 months of seizure freedom before driving a private vehicle,¹³ while most US states require a median of 6 months.¹⁴ Many jurisdictions allow patients with PSE with ongoing sleep seizures to drive, but the definition of PSE varies.⁶ In Australia and parts of Europe, patients may return to driving after 12 months of sleep seizures with no previous awake seizures.^{13,15} New Zealand defines PSE after 3 years,¹⁶ while Japan uses 24 months.¹⁷ In the United States, 15 of the 50 states have less stringent restrictions for patients with nocturnal seizures.⁶ None of these guidelines specify the number of consecutive sleep-only seizures required to define PSE.

Determining when patients with PSE should be allowed to drive a private motor vehicle depends on the relative risk of a

Figure 2 Cumulative Risk of Seizure Recurrence After a First-Ever Unprovoked Seizure From Sleep

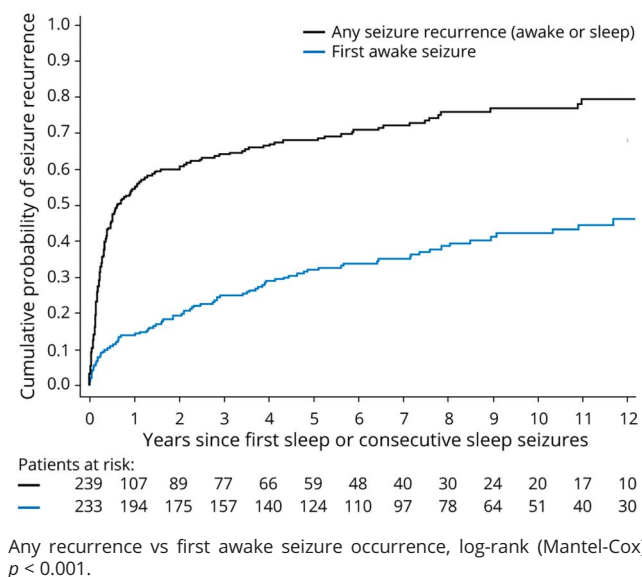
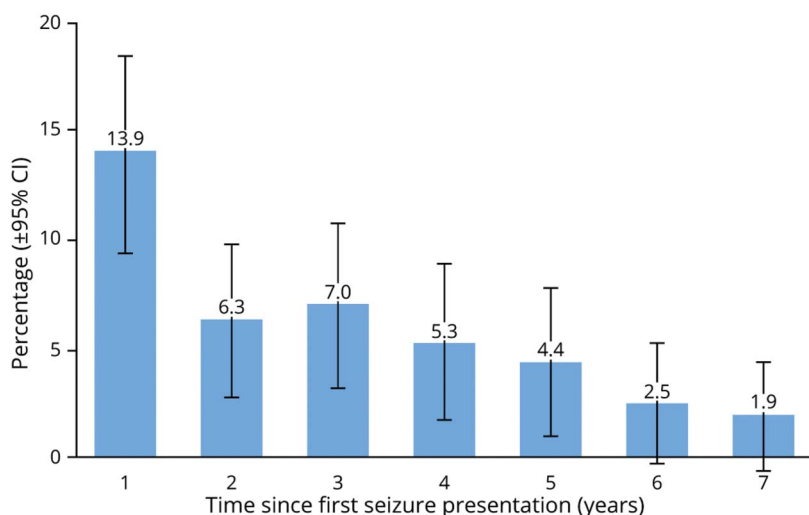


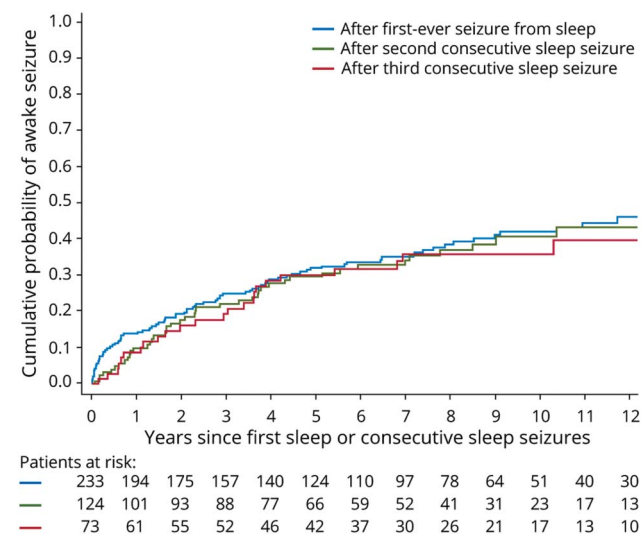
Figure 3 Annual Conditional Probability of First Awake Seizure After a First-Ever Unprovoked Seizure From Sleep (95% CI)



crash that the community is willing to accept. No patient with epilepsy will ever be completely risk-free of awake seizures. According to a Belgium Traffic Bureau study, people with conditions like epilepsy may be allowed to drive when their accident rate ratio (relative risk of a crash) falls to ≤ 2 .¹⁸ This relative risk is comparable to the increased crash rate accepted for many drivers, including those driving when sleep deprived or within legal alcohol limits.¹⁸ An accident rate ratio of 2 for epilepsy equates to the annual chance of an awake seizure of 20% (i.e., a monthly risk consistently below 1.7%) based on a daily driving time of 1 hour and other factors.^{18,19} The time at which the monthly risk (conditional probability) of an awake

seizure consistently fell below 1.7% per month was 8 months after the first-ever unprovoked seizure from sleep, 3 months after the second consecutive sleep seizure, and 0 months after the third consecutive sleep seizure (76 patients, with a median time from the first to third sleep seizure of 8.6 months). Therefore, our data provide support for jurisdictions that allow a return to driving a private motor vehicle after 12 months of sleep-only seizures with no previous awake seizures. However, other unaccounted-for factors influence driving safety, including daytime sleepiness after seizures, adverse effects of the underlying cause, and ASMs.^{19,20}

Figure 4 Cumulative Probability of First Awake Seizure After First-Ever Unprovoked Seizure From Sleep or 2 or 3 Consecutive Seizures From Sleep (χ^2 $p \leq 0.001$)



This prospective study assessed the risk of PSE and awake seizures after a first-ever unprovoked sleep seizure, with a larger sample size and longer follow-up than previously.^{1,4,5} However, our study has several limitations. First, common to all first seizure studies, the reported first event may not truly be the first-ever seizure. Seizures from sleep are more likely to be unrecognized, particularly sleep-related focal seizures and those in patients sleeping alone. Ninety-two percent of our patients had a witness account, and most had a physical manifestation to alert them to their seizure. Second, the median time from first seizure to review was 1 month. This delay may have influenced identification of early seizure recurrence. Inclusion of patients with early seizure recurrence while awaiting assessment may have overestimated seizure recurrence, whereas their exclusion would have underestimated it, with both approaches having time of ascertainment bias.²¹ However, our data show similar findings with either approach.²² The rate of recurrent focal sleep seizures was low, raising the possibility of unrecognized or unreported focal events. More extensive assessments such as multiday ambulatory EEGs or implanted EEG recording devices would improve seizure detection rate but are impractical for most patients.^{23,24} In many patients, the electroclinical syndrome was unclassified because it may be difficult for bed partners to

Table 1 Clinical and Investigation Findings in Patients With First-Ever Unprovoked Seizure From Sleep, Followed by Either Pure Sleep Epilepsy or Subsequent Awake Seizure

	Pure sleep epilepsy (n = 85)	Subsequent awake seizure (n = 89)	p Value
Mean duration of follow-up since second seizure (range), y	7.6 (0.2–15.8)	8.4 (0.1–17.7)	0.19
Sex male, n (%)	56 (66)	50 (56)	0.21
Age, mean (range), y	43.0 (14–82)	42.7 (15–81)	0.98
Seizure type (of initial seizure), n (%)			0.51
Tonic-clonic	83 (97.6)	87 (97.7)	
Focal unaware	2 (2.4)	1 (1.1)	
Focal aware	0 (0)	1 (1.1)	
Electroclinical syndrome, n (%)			0.01 ^a
Generalized	0	5 (5.6)	
Focal	38 (44.7)	52 (58.4)	
Unclassified	47 (55.3)	32 (36.0)	
Family history of seizures	7 (8.2)	16 (18.6)	0.07
Etiology, n (%)			>0.99
Remote symptomatic	22 (25.9)	23 (25.8)	
Idiopathic	63 (74.1)	66 (74.2)	
Remote symptomatic etiology subgroups, n (%)			0.24
Stroke (clinical)	2 (9.1)	5 (21.7)	
Head injury	6 (27.3)	2 (8.7)	
Tumor	2 (9.1)	6 (26.1)	
Other	12 (54.5)	10 (43.5)	
Epileptogenic lesion on imaging	17 (20)	22 (24.7)	0.47
EEG, n (%)			0.06 0.03 ^a 0.38
No epileptiform abnormalities	74 (89.2)	69 (77.5)	
Epileptiform abnormalities	9 (10.8)	20 (22.5)	
Generalized	0 (0)	6 (6.7)	
Focal	9 (10.8)	14 (15.7)	
Time to second seizure, mean (range), y	1 (1 day–11 years)	1.5 (1 day–17.6 y)	0.87
Treated immediately after first seizure, n (%)	14 (16.5)	29 (32.6)	0.01 ^a
Cumulative risk of death, % (95% CI)			0.85
1 y	0.02 (0–0.05)	0.00 (0)	
2 y	0.02 (0–0.05)	0.00 (0)	
5 y	0.06 (0–0.12)	0.06 (0.01–0.11)	

^a Not significant after correction for multiple comparisons.

differentiate generalized-onset tonic-clonic seizures from focal to bilateral tonic-clonic or focal hypermotor seizures. Investigations did not clarify this in many patients. Furthermore,

the high rate of tonic-clonic seizures may reflect a failure to identify focal-onset seizures, given the propensity for focal seizures to propagate to bilateral tonic-clonic seizures during

Table 2 Univariate HR for Awake Seizure Occurrence

	HR (95% CI)	p Value
Age	1.0 (0.99–1.01)	0.96
Family history epilepsy	2.04 (1.18–3.52)	0.01
Seizure type (tonic-clonic vs others)	0.88 (0.12–6.31)	0.89
Electroclinical syndrome (vs unclassified)		
Generalized	3.02 (1.17–7.78)	0.02
Focal	1.59 (1.02–2.49)	0.04
Epileptogenic lesion on imaging	1.12 (0.69–1.82)	0.64
Epileptiform abnormalities on EEG	1.75 (1.06–2.89)	0.027
Focal epileptiform EEG	1.44 (0.81–2.56)	0.21
Generalized epileptiform EEG	2.56 (1.11–5.88)	0.026
Remote symptomatic etiology	0.93 (0.56–1.45)	0.67
Treated after first seizure	1.43 (0.92–2.24)	0.11
Time to recurrence	1.00 (1.00–1.00)	0.21

Abbreviation: HR = hazard ratio.

sleep.²⁵ This may relate to enhanced cortical network synchrony during non-REM sleep or changes in neurotransmitters during sleep.^{25,26} Fifty-seven of our patients (24%) underwent only CT head imaging, reducing the detection of lesions that may have clarified the electroclinical diagnosis. While some had contraindications to MRI, 20 of these 57 patients showed definite epileptogenic lesions on CT, so a relevant lesion may have been missed without MRI in only 37 patients (15%). Last, our patients were seen at tertiary hospital-based specialist clinics, and recruitment was predominantly through emergency department referrals rather than community-based settings. While this may lead to selection bias, our study population was broadly inclusive, including the relatively high proportion of remote symptomatic seizures, and is similar to other prospective observational studies in adults.^{21,27} All investigations and management decisions were based on clinical indications. Therefore, our findings are applicable to patients with a first-ever seizure encountered in clinical practice.

We have prospectively assessed the evolution of PSE after a first-ever seizure from sleep and suggest criteria to define PSE. Although PSE can be identified after 3 years of sleep-only seizures, a 2% to 5% ongoing annual risk of seizures from wakefulness remains long term. Our findings may be used to inform driving guidelines for patients with sleep-only seizures.

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Nicholas Lawn, MBBS, FRACP	Western Australian Adult Epilepsy Service, Perth, Australia	Drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data
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